Ship Detection by an Airborne Passive Interferometric Microwave Sensor (PIMS)

Hailiang Lu, Yinan Li, Hao Li, Rongchuan Lv, Liang Lang, Qingxia Li, Guangnan Song, Pengfei Li, Ke Wang, Liang Xue, and Dong Zhu, *Member, IEEE*

***Abstract*— Ship detection is important for a wide range of applications. In this article, an airborne passive interferometric microwave sensor (PIMS) is proposed as a powerful complementary tool for ship detection. The fundamental of ship detection by the airborne PIMS is introduced. A specification called "the detection factor" is addressed to quantitatively assess the performance of the airborne PIMS for ship detection. The detection probability is discussed and addressed based on the constant false-alarm rate (CFAR). Numerical simulations are performed to demonstrate the feasibility of ship detection by the airborne PIMS. The impacts of the atmosphere and the sea wind on the performance of the airborne PIMS for ship detection are also addressed in terms of theoretical analyses and numerical simulations. The airborne experiments are also carried out by an X-band PIMS to verify the practicability of ship detection by the airborne PIMS in a sunny weather and a cloudy weather. The data processing procedure and a ship detection algorithm in the experiments are introduced. The airborne experimental results reveal that the ships can be effectively detected and tracked by the airborne X-band PIMS in a sunny weather and a cloudy weather. It demonstrates the practicability of ship detection by the airborne PIMS. The ship detection probability is also analyzed, which agrees well with the theoretical detection probability.**

***Index Terms*— Airborne, interferometric, microwave sensor, passive, ship detection.**

I. Introduction

SHIP detection is vitally important for a wide range of applications, including illegal smuggling detection, traffic surveillance and management, fishery management, oil-spill detection, and naval warfare [1], [2]. The current ship detection is mainly based on the airborne/spaceborne synthetic aperture radar (SAR) [3], [4], high-frequency surface wave radar (HFSWR) [5]–[7], automatic identification system (AIS) [8], [9], airborne/spaceborne optical sensors [10], and infrared sensor [11], [12]. The HFSWR has the ability to detect and track ships in real time and beyond the horizon continuously [13]–[15]. However, its coarse spatial resolution leads to poor position accuracy for ships, and sea clutter, as well as interference and external noise, creates a severe disturbance during ship detection [7]. The AIS is mandatory for a portion of traffic and can provide a wealth of data for Maritime Situational Awareness (MSA) [16]. However, the AIS cannot detect noncooperative ships [9], [13]. Optical sensors are more suitable for ship detection in good weather conditions due to higher resolution. However, optical sensors cannot perform well at night and usually suffer from interference by low visibility conditions, such as clouds or mist, or from ocean waves [4], [17]. Infrared sensors can perform well at night. They are also able to penetrate mist and fog to a far greater extent than the optical sensors [18]. However, infrared sensors also have difficulty when trying to image through clouds or slight drizzle [18].

Unlike the optical and infrared sensors, microwave sensors have imaging capability through low visibility conditions, such as haze, fog, snow, light rain, dust clouds, and smoke screens [19]. Microwave sensors include active sensors and passive sensors. SAR is a main active microwave sensor for ship detection [20]. However, SAR usually suffers from sea clutter and high-level speckles and is insensitive to the objects with small radar signatures (such as wooden materials) and is difficult for humans to interpret [21], [22]. Compared with active microwave sensors, the radiometric systems embedding passive microwave sensors have no speckle issue and suffer little from sea clutter, which seem to be more efficient, difficult to unveil, and portable [23]. In addition, the radiometric systems are also sensitive to objects with small radar signatures, such as wooden materials and fiberglass materials [19]. It is clear that these aforementioned sensors have their own advantages and weaknesses. They can complement each other for ship detection in all-weather and all-day conditions.

Passive microwave sensors have been known for many years. Studies carried out have clearly demonstrated the utility of the imagery acquired, such as all-weather ground-based surveillance, landing aids for civilian aircraft, and ground-movement control at civil airports [18], [24].

Manuscript received August 25, 2019; revised October 26, 2019; accepted November 4, 2019. Date of publication December 18, 2019; date of current version March 25, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 41706204 and Grant 61901244, in part by the Young Innovation Foundation of XueSen Qian, and in part by the Independent Investigate Project of the China Academy of Space Technology (Xi'an) under Grant Y17-KJCX-04. *(Corresponding author: Dong Zhu.)*

H. Lu, Y. Li, H. Li, R. Lv, G. Song, P. Li, K. Wang, and L. Xue are with the China Academy of Space Technology (Xi'an), Xi'an 710100, China (e-mail: 396689lhl@163.com).

L. Lang and Q. Li are with the School of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: l_lang@mail.hust.edu.cn).

D. Zhu is with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: zhudong@mail.tsinghua.edu.cn).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2953355

0196-2892 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

The great advantage of a passive microwave sensor is that it can be used not only in the day and night conditions but also in low visibility conditions that normally limit the "seeing" ability of both optical and infrared sensors [19]. However, the spatial resolution of passive microwave radiometric imaging is a very important issue in a variety of applications, especially for airborne and spaceborne passive imaging sensors [25], [26]. Traditional mechanic scanning microwave radiometers usually cannot attain the demanded resolution even when working at microwave because of strict bulk and weight limit. To that end, the synthesis aperture interferometric radiometer (SAIR) has been proposed in 1980s, which can realize passive imaging with high resolutions [25], such as electronically steered thinned array radiometer (ESTAR), microwave imaging radiometer using aperture synthesis (MIRAS), and passive advanced unit-synthetic aperture (PAU-SA) [27].

With the efficient imaging capability in all-weather and all-day conditions, an airborne SAIR, the passive interferometric microwave sensor (PIMS), is proposed as a powerful complementary tool for ship detection in this article. In this article, the detection factor is defined to quantitatively assess the performance of the PIMS for ship detection, the impacts of the atmosphere and the sea wind are addressed, and the airborne experiments are also performed to verify ship detection by the airborne PIMS.

The rest of this article is organized as follows. In Section II, the fundamental of ship detection by the airborne PIMS is introduced. The main specifications of the PIMS are addressed in Section III. The impacts of the atmosphere and the sea wind are discussed in Section IV. In Section V, the airborne experiments are performed. Finally, the conclusion is given in Section VI.

II. Fundamental of Ship Detection by the Airborne PIMS

According to Plank's blackbody radiation law, all matter radiates electromagnetic energy. The radiation is a consequence of the interaction between the atoms and molecules in materials [28]. The brightness temperature (TB) image by the passive microwave sensors is the map of the electromagnetic energy of the matter in microwave. The passive microwave sensors are microwave radiometers, which are with high gain, high stability, and high sensitivity receivers operating at microwave [29]. The object's surface TB image is the product of the emissivity ε of the object's physical thermodynamic temperature T_0

$$T_B = \varepsilon \cdot T_0. \tag{1}$$

A perfect radiator (absorber) has $\varepsilon = 1$ and is known as a blackbody. A perfect reflector (nonabsorber) has $\varepsilon = 0$, such as a metallic object. The emissivity of an object, which is polarization-dependent, is a function of the dielectric properties of its constituents, its surface roughness, and the angle of observation. The artificial objects and natural matters usually exhibit different emissivities in microwave, which leads to a high contrast existing in the passive microwave images. Thus, the target can be detected by the high contrast in the passive microwave images.

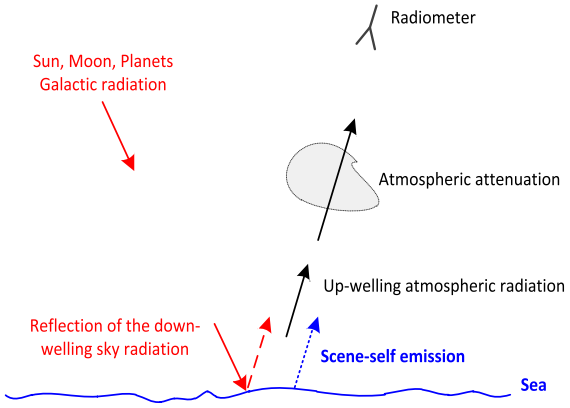


Fig. 1. Contributions of the observed scene TB from an airborne PIMS. The red solid line denotes the downwelling sky radiation, the black solid line denotes the upwelling atmospheric radiation between the scene and the observer, the red dashed line denotes the reflection of the downwelling sky radiation by the sea scene, and the blue dashed line denotes the emission from sea scene-self.

From the air, the observed TB of the scene includes three main contributions: the emission from scene-self, the reflection of the downwelling sky radiation by the scene, and the upwelling atmospheric radiation between the scene and the observer, as shown in Fig. 1 [19]. In addition, reflection from parts of the ship that comes from the sky radiation closer to the horizon also has a contribution to the observed TB of the scene [28]. The propagation of electromagnetic energy from the scene to the observer always suffers from atmospheric attenuations (see Fig. 1) [19]. In addition, the downwelling sky radiation, the upwelling atmospheric radiation, and atmospheric attenuations exhibit discrepancies in frequency, such as 10.7, 18.7, 23.8, 37, and 89 GHz.

The ships are usually made of painted metals. The emissivity of the painted metal does not exceed 0.1 below 100 GHz [19], and the reflection TB from the sky is about 2.7–60 K below 100 GHz [19]. Thus, the painted metallic ship is a "low" TB object from the air. However, the emissivity of the sea background is about 0.4–0.7 [19], and the sea TB is warm about 120–210 K below 100 GHz. As shown in Fig. 2 (left), the original TB maps of a metallic ship at sea with 500-m altitude are given at 10.7, 18.7, 23.8, 37, and 89 GHz. As shown in Fig. 2 (right), the simulated TB maps of a metallic ship at sea, which are obtained by an airborne passive microwave sensor with 500-m altitude, are also given at 10.7, 18.7, 23.8, 37, and 89 GHz. Numerical simulations performed in Fig. 2 are based on the ray-tracing scene simulation model [28]. Therefore, the reflection of the downwelling sky radiation by the scene, the up-welling atmospheric radiation between the scene and the observer, and the atmospheric attenuations are considered in Fig. 2. In addition, the contribution of reflections from the sky radiation closer to the horizon by the parts of the ship is also considered in Fig. 2, such as the sides of the ship.

As shown in Fig. 2, it is found that the metallic ship shows a high contrast in the passive interferometric microwave image (PIMI) and exhibits a "cold" TB characteristic by contrast with the sea background from the air, as expected. As a result, the ships can be detected by the airborne PIMS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

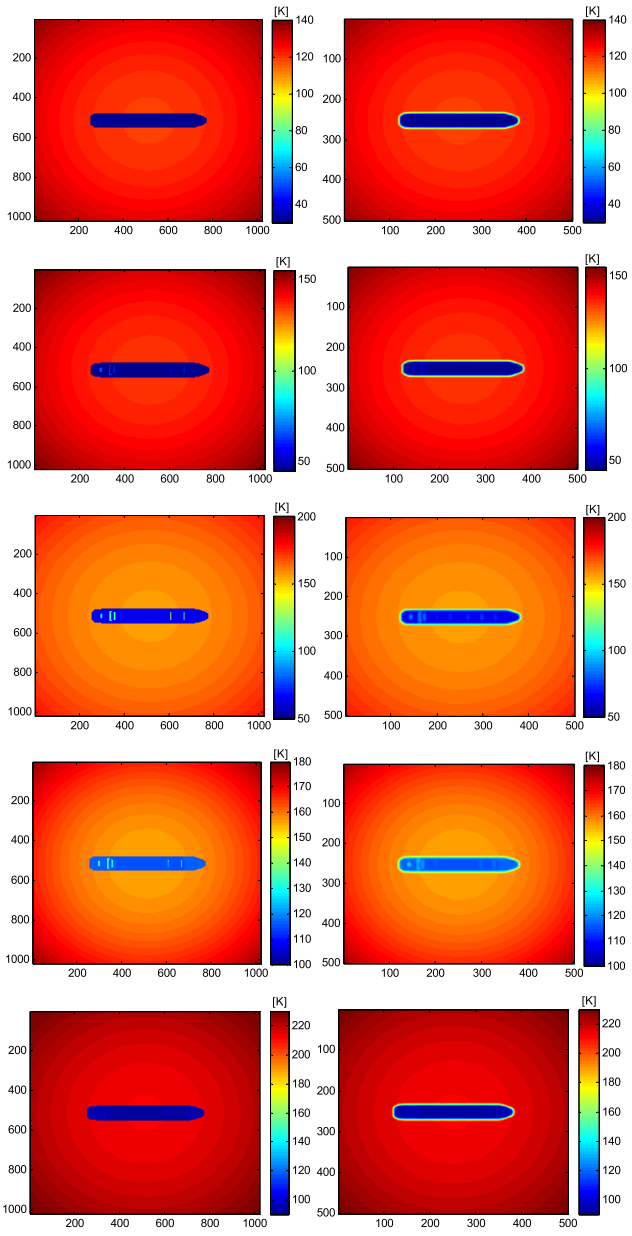


Fig. 2. (Left) Original apparent TB maps of a metallic ship with 500-m altitude at 10.7, 18.7, 23.8, 37, and 89 GHz. (Right) Simulated TB maps of a metallic ship at 10.7, 18.7, 23.8, 37, and 89 GHz, which are obtained by an airborne PIMS at 500-m height.

in terms of the "cold" TB characteristic of the ships in the PIMI.

III. Main Specifications of PIMS

In the PIMS devoted to the remote sensing of the Earth, the system sensitivity and the angular resolution are two key specifications, such as in MIRAS [30]. In this section, a new specification called "the detection factor" is defined and addressed to quantitatively assess the performance of an airborne PIMS for ship detection. The detection probability is addressed for ship detection by an airborne PIMS. Numerical simulations are also performed to analyze the feasibility of ship detection by the airborne PIMS.

A. Detection Factor of Airborne PIMS

For an airborne PIMS, the spatial resolution in the x-axis (R_X) and the y-axis (R_Y) can be calculated as

$$R_X = L \cdot \Delta\theta_X \tag{2}$$

$$R_Y = L \cdot \Delta\theta_Y \tag{3}$$

where $\Delta\theta_X$ and $\Delta\theta_Y$ denote the x- and y-axes angular resolution (in radian) of the airborne PIMS, respectively. L is the distance between the ship and the airborne PIMS.

At the same time, the TB change ΔT in one pixel by the ship can be estimated as

$$\Delta T = \frac{\Delta T_{\text{target}} \cdot S_{\text{eff}}}{R_X \cdot R_Y} \tag{4}$$

where $\Delta T_{\text{target}} = |T^{\text{ap}}_{\text{target}} - T^{\text{ap}}_{\text{sea}}|$ is the antenna apparent TB absolute difference between the ships and the sea background and S_{eff} denotes the microwave radiation cross section of the ships in the pixel.

Suppose the length and the width of the ship are L_S and W_S ($L_S > W_S$), and the spatial resolution in the x- and y-axes is equal ($R_X = R_Y = R$). Three conditions are considered in the PIMI for ships.

When $W_S < L_S < R_X = R_Y = R$, the ship is considered as a point target in the PIMI. For simplify, the microwave radiation cross section in the pixel of the ships is approximated as

$$S_{\text{eff}} \approx L_S \cdot W_S. \tag{5}$$

When $L_s > R_X = R_Y = R$ and $W_s < R_X = R_Y = R$, the ship is considered as a line target in the PIMI. In this case, the microwave radiation cross section in the pixel of the ships is complicated. To simplify, S_{eff} can also be approximated as

$$S_{\text{eff}} \approx W_S \cdot \max(R_X, R_Y) = W_S \cdot R. \tag{6}$$

When $L_s > R_X = R_Y = R$ and $W_S > R_X = R_Y = R$, the ship is considered as an area target in the PIMI. In this case, the microwave radiation cross section in the pixel of the ships is approximated as

$$S_{\text{eff}} \approx R_X \cdot R_Y = R^2. \tag{7}$$

Then, by substituting (2) and (3) into (4), the TB change ΔT is given by

$$\Delta T = \frac{\Delta T_{\text{target}} \cdot S_{\text{eff}}}{L^2 \cdot \Delta\theta_X \cdot \Delta\theta_Y}. \tag{8}$$

When ΔT is not smaller than the sensitivity ΔT_B of the PIMS

$$\Delta T \geq \Delta T_B. \tag{9}$$

The ship is considered to be detectable, and (8) is substituted into (9)

$$\frac{1}{\Delta T_B \cdot \Delta\theta_X \cdot \Delta\theta_Y} \geq \frac{L^2}{\Delta T_{\text{target}} \cdot S_{\text{eff}}}. \tag{10}$$

According to (10), the reciprocal of the product of the sensitivity and the angular resolution can be considered to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

quantitatively assess the performance of an airborne PIMS for ship detection [31]–[33]. Thus, the detection factor is defined as

$$Df_{\text{system}} = \frac{1}{\Delta T_B \cdot \Delta\theta_X \cdot \Delta\theta_Y}. \tag{11}$$

For ships, the needed detection factor of the targets is also defined as

$$Df_{\text{target}} = \frac{L^2}{\Delta T_{\text{target}} \cdot S_{\text{eff}}}. \tag{12}$$

When Df_{system} is not smaller than Df_{target}, the ships are considered to be detectable [34].

According to (10)–(12), the detection performance of the airborne PIMS for ship detection can be assessed quantitatively by the detection factor, which is defined as the reciprocal of the product of the sensitivity and the angular resolutions of the PIMS. Moreover, the larger the detection factor, the better the detection performance of the airborne PIMS.

In addition, a ship in the PIMI varies from an area target to a line target and a point target as the distance increases. To detect the ship as far as possible, the ship is usually a line target or a point target in the PIMI. As a result, this article mainly focuses on the ships as a line target or a point target in the PIMI by the airborne PIMS.

B. Detection Probability of Airborne PIMS

In SARs, the detection probability is usually used to assess the performance of an SAR. Thus, the detection probability is also considered to assess the performance of the airborne PIMS for ship detection.

1) Detection Probability: In the PIMI with ships, the signal-to-noise ratio (SNR) in one pixel associated with ships is defined as the ratio of the TB change ΔT by the ship to the sensitivity ΔT_B as follows:

$$\text{SNR} = \frac{\Delta T}{\Delta T_B}. \tag{13}$$

From (8), (11), and (12), the ratio of the TB change ΔT and the sensitivity ΔT_B can be estimated by

$$\frac{\Delta T}{\Delta T_B} = \frac{Df_{\text{system}}}{Df_{\text{target}}}. \tag{14}$$

In the PIMI, the microwave TB in each pixel is stochastic and follows a Gaussian distribution. As the red line shown in Fig. 3, the microwave TB distribution of the sea background in the PIMI is given by

$$p(T) = \frac{1}{\sqrt{2\pi}\,\Delta T_B} \exp\left(-\frac{(T - T_0)^2}{2\Delta T_B^2}\right) \tag{15}$$

where T denotes the TB in a pixel and T_0 is the TB mean of the pixel. When there is a ship in a pixel, the TB mean in the pixel is

$$T_t = T_0 + \Delta T. \tag{16}$$

In this case, as the blue line shown in Fig. 3, the TB distribution in a pixel is

$$p(\tilde{T}) = \frac{1}{\sqrt{2\pi}\,\Delta T_B} \exp\left(-\frac{(\tilde{T} - T_t)^2}{2\Delta T_B^2}\right). \tag{17}$$

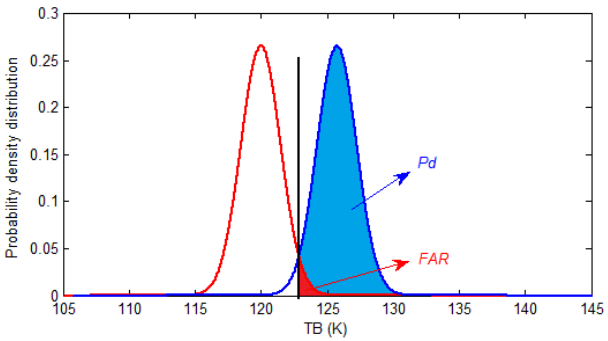


Fig. 3. Sketch of detection probability and FAR. The red line denotes the microwave TB distribution of the sea background, the blue line denotes the microwave TB distribution of a ship in a pixel, and the black line denotes the detection threshold.

Since the sea background is a "warm" TB scene, the TB change by a maritime is negative when it is a "low" TB object (such as metallic ships), and the TB change by a maritime is positive when it is a "high" TB object (such as an island). Although the integral ranges of the detection probability and false-alarm rate (FAR) are different in a "low" TB object and a "high" TB object, the integral ranges are symmetrical when the absolute TB changes ΔT are equal. As a result, the detection probability and the FAR are the same in a "low" TB object and a "high" TB object when the absolute TB changes are equal.

Suppose that the TB change ΔT is positive and the detection threshold is $T_0 + \Delta T_{\text{threshold}}$, as the black line shown in Fig. 3, the FAR is calculated as

$$\begin{aligned} P_f &= \int_{T_{\text{threshold}}}^{+\infty} p(T)dT \\ &= \int_{\Delta T_{\text{threshold}}}^{+\infty} \frac{1}{\sqrt{2\pi}\,\Delta T_B} \exp\left(-\frac{(\Delta T_{\text{threshold}})^2}{2\Delta T_B^2}\right) dT. \end{aligned} \tag{18}$$

The detection probability P_d can also be calculated as

$$\begin{aligned} P_d = \int_{T_{\text{threshold}}}^{+\infty} p(\tilde{T})dT = \int_{\Delta T_{\text{threshold}}}^{+\infty} \frac{1}{\sqrt{2\pi}\,\Delta T_B} \\ \times \exp\left(-\frac{(\Delta T_{\text{threshold}} - \Delta T)^2}{2\Delta T_B^2}\right) dT. \end{aligned} \tag{19}$$

Equation (18) is a standard Gaussian distribution function, and FAR(P_f) can also be written as

$$P_f = Q\left(\frac{\Delta T_{\text{threshold}}}{\Delta T_B}\right). \tag{20}$$

Then

$$\frac{\Delta T_{\text{threshold}}}{\Delta T_B} = Q^{-1}(P_f). \tag{21}$$

Similarly, (19) is also written as

$$\begin{aligned} P_d &= Q\left(\frac{\Delta T_{\text{threshold}} - \Delta T_{B_\text{seff}}}{\Delta T_B}\right) \\ &= Q\left(\frac{\Delta T_{\text{threshold}}}{\Delta T_B} - \text{SNR}\right) \end{aligned} \tag{22}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

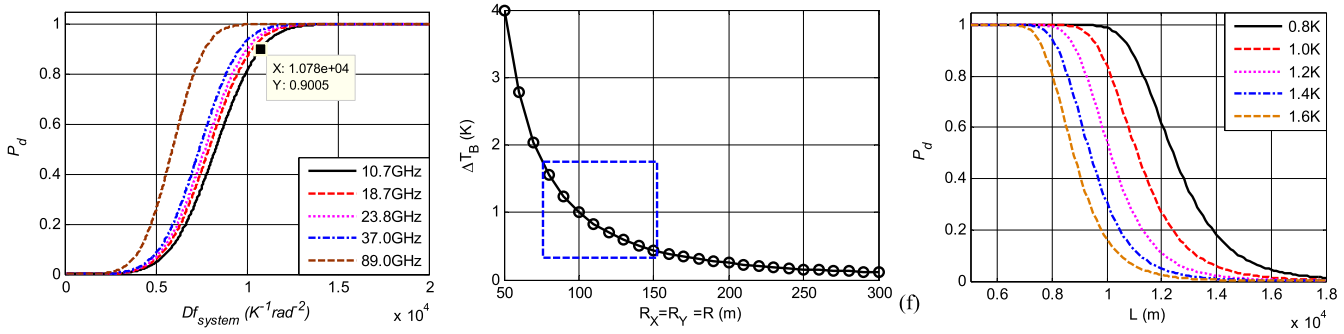


Fig. 4. (Left) Detection probability as a function of the detection factor of an airborne PIMS, which operates at 10.7, 18.7, 23.8, 37, and 89 GHz. (Middle) Sensitivity as a function of the spatial resolution ($R_X = R_Y = R$) with $Df_{\text{system}} = 10^4\,K^{-1}rad^{-2}$ and $L = 10$ km. (Right) Detection probability as a function of the detection distance in 37 GHz.

and

$$\frac{\Delta T_{\text{threshold}}}{\Delta T_B} - \text{SNR} = Q^{-1}(P_d). \tag{23}$$

By substituting (21) into (23), the SNR can be expressed as

$$\text{SNR} = Q^{-1}(P_f) - Q^{-1}(P_d). \tag{24}$$

According to (24), SNR is a function of the detection probability and FAR. Thus, (21) can be expressed as

$$\text{ROC}(P_d, P_f) = Q^{-1}(P_f) - Q^{-1}(P_d). \tag{25}$$

By substituting (8) into (13), the SNR can be estimated by

$$\text{SNR} = \frac{1}{\Delta T_B \cdot \Delta\theta_X \cdot \Delta\theta_Y} \cdot \frac{\Delta T_{\text{target}} \cdot S_{\text{eff}}}{L^2}. \tag{26}$$

From (24)–(26), the detection probability and FAR can also be expressed as

$$\text{ROC}(P_d, P_f) = \frac{1}{\Delta T_B \cdot \Delta\theta_X \cdot \Delta\theta_Y} \cdot \frac{\Delta T_{\text{target}} \cdot S_{\text{eff}}}{L^2}. \tag{27}$$

In (27), the detection probability and the FAR are derived. The detection probability can be established at the constant FAR (CFAR) when the parameters in the left-hand side of (27) are known. According to (27), the detection probability depends on the FAR, the sensitivity and angle resolutions of the airborne PIMS, the microwave radiation cross section of the ships, and the TB absolute difference between the ships and the sea background. In addition, the detection probability also depends on the detection distance (L).

2) Detection Distance: The detection distance is also considered to assess the performance of a sensor for ship detection. According to (27), the detection distance (L) of an airborne PIMS for a ship can be derived as

$$L = \sqrt{\frac{1}{\Delta T_B \cdot \Delta\theta_X \cdot \Delta\theta_Y} \cdot \frac{\Delta T_{\text{target}} \cdot S_{\text{eff}}}{\text{ROC}(P_d, P_f)}}. \tag{28}$$

According to (28), the detection distance (L) can be estimated when the parameters associated with the ships and the PIMS in (28) at a constant detection probability (P_d) and FAR (P_f).

It must be pointed out that the parameters in (27) and (28) are the function of the direction cosines ($\xi = \sin(\theta)\cos(\phi)$ and $\eta = \sin(\theta)\sin(\phi)$) in the PIMI. From (27), the detection probability and the FAR in the pixel (ξ, η) can be expressed as

$$\text{ROC}(P_d, P_f) = \frac{\Delta T_{\text{target}}(\xi,\eta) \cdot S_{\text{eff}}(\xi,\eta)}{\Delta T_B(\xi,\eta) \cdot \Delta\theta_X(\xi,\eta) \cdot \Delta\theta_Y(\xi,\eta)} \cdot \frac{1}{L^2}. \tag{29}$$

As a result, the detection probability of the pixels within the alias-free field of views of the airborne PIMS can be estimated by (29).

C. Numerical Simulations

Numerical simulations are performed to assess the feasibility of ship detection by the airborne PIMS. In the numerical simulations, the sea surface salinity is 32 psu, the sea surface temperature is 300 K, and the ship is metallic with 50-m length and 10-m width, which is considered as a point target in the numerical simulations. Assume that the weather is good and the atmospheric attenuation is neglected. In the numerical simulations, the detection probability with $P_f = 10^{-5}$ is analyzed in the pixel ($\xi = 0, \eta = 0$).

According to (29), the detection probability P_d as a function of the detection factor of the airborne PIMS, which operates at 10.7, 18.7, 23.8, 37, and 89 GHz, is shown in Fig. 4 (left). It is found that the detection probability increases as the detection factor Df_{system} increases, as expected. It also means that the detection factor, defined in (11), is effective to quantitatively assess the performance of the airborne PIMS for ship detection. It is also found that the PIMS in high frequency shows a better detection probability without considering atmospheric attenuation, which is caused by higher ΔT_{target} in high frequency. In addition, it can also be seen that the detection probability P_d exceeds 80% with $Df_{\text{system}} > 10^4 K^{-1}\text{rad}^{-2}$.

According to (11), the detection factor Df_{system} can also be expressed as

$$Df_{\text{system}} = \frac{L^2}{\Delta T_B \cdot R_X \cdot R_Y}. \tag{30}$$

Thus, the sensitivity as a function of the spatial resolution ($R_X = R_Y$) is shown in Fig. 4 (middle) when $Df_{\text{system}} = 10^4\,K^{-1}\text{rad}^{-2}$ and $L = 10$ km. It is found that the sensitivity increases as the spatial resolution decreases. It can be seen

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

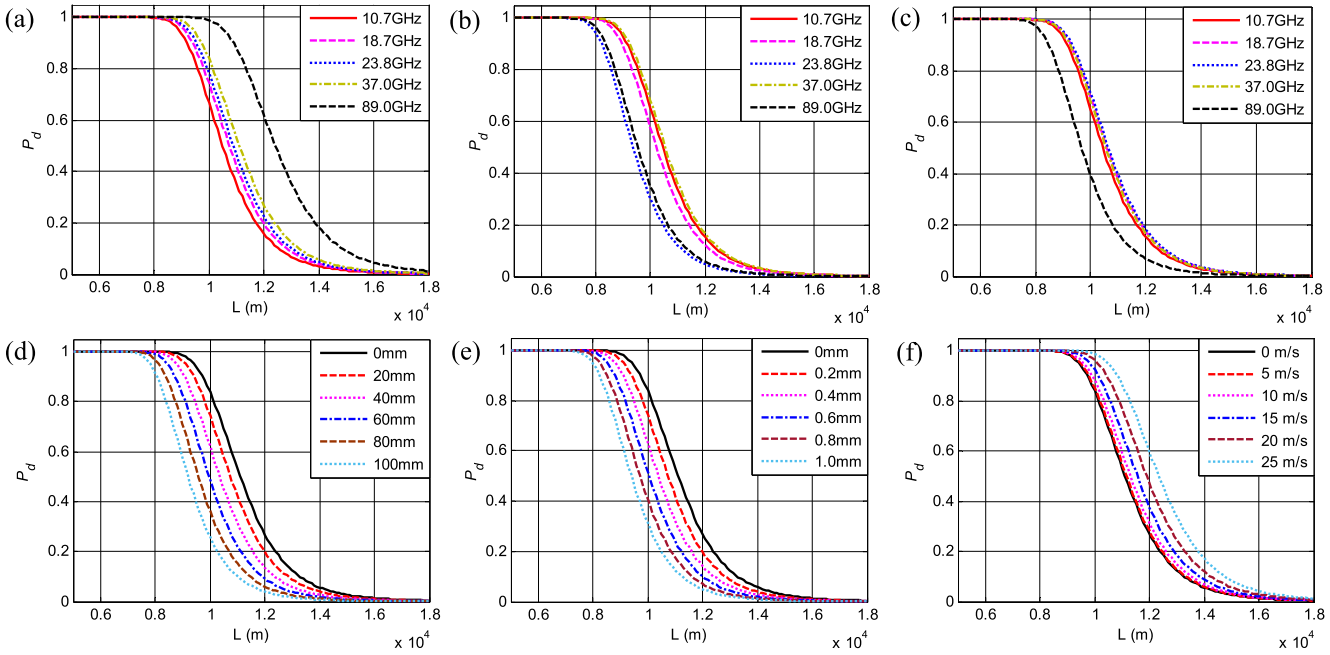


Fig. 5. Detection probability P_d with $P_f = 10^{-5}$ as a function of the detection distance L with (a) ideal case $\tau = 1$, (b) 30-mm vapor, and (c) 0.3-mm water in 10.7, 18.7, 23.8, 37.0, and 89.0 GHz, and the detection probability P_d with $P_f = 10^{-5}$ as a function of the detection distance with (d) different vapor contents, (e) different water contents, and (f) different wind speeds in 37 GHz.

that the poorer the spatial resolution, the higher the sensitivity as needed to achieve the expected detection factor. In fact, the spatial resolution better than 50 m or the sensitivity better than 0.2 K is very difficult to be achieved by the airborne PIMS with $L = 10$ km. As a result, a compromise should be made between the spatial resolution and sensitivity, as the blue dotted rectangle shown in Fig. 4 (middle).

According to (28), the detection probability as a function of the detection distance L in 37 GHz is shown in Fig. 4 (right) when the angular resolution is 0.5° and the sensitivity is 0.8, 1.0, 1.2, 1.4, and 1.6 K. The result exhibits that the detection probability increases as the detection distance decreases, as expected. It can also be seen that the metallic ship can be detected with $P_d \geq 99\%$ by an airborne PIMS in 37 GHz with $\Delta\theta_X = \Delta\theta_Y = 0.5°$ and $\Delta T_B = 0.8$ K when the detection distance L is less than 10 km. In this case, the antenna synthesis aperture of the airborne PIMS in 37 GHz is about 1 m, which is practicable. This indicates that the ships at sea detected by the airborne PIMS are feasible.

IV. Impacts of the Atmosphere and the Sea Wind

It is well known that the attenuation of the atmosphere cannot be neglected in bad weather due to high water and vapor contents [35], such as moderate rain and heavy rain. Bad weather can reduce the apparent TB absolute difference ΔT_{target} in (27). On the other hand, the sea wind can change the sea surface roughness to increase the microwave TB of sea background [36], and it also leads to change the apparent TB absolute difference ΔT_{target} in (27). As a result, the atmosphere and sea wind will have a nonnegligible impact on the performance of the airborne PIMS for ship detection. Hence, the impacts of the atmosphere and the sea wind are also addressed in terms of theoretical analyses and numerical simulations in this section.

A. Impact of the Atmosphere on PIMS

When considering the impact of the atmosphere, ΔT_{target} can be expressed as

$$\Delta T_{\text{target}} = |T_{\text{target}} - T_{\text{sea}}| \cdot \tau \tag{31}$$

where τ is the transmittance of the atmosphere in one pixel, and T_{target} and T_{sea} are the atmosphere bottom TB of the ship and the sea background in one pixel, respectively.

In this case, (27) can be further expressed as

$$\text{ROC}(P_f, P_d) = \frac{1}{\Delta T_B \cdot \Delta\theta_X \cdot \Delta\theta_Y} \cdot \frac{S_{\text{eff}} |T_{\text{target}} - T_{\text{sea}}| \cdot \tau}{L^2}. \tag{32}$$

In the ideal case that the atmospheric attenuation is neglected, the transmittance of the atmosphere is equal to $1(\tau = 1)$. In fact, the atmospheric attenuation cannot be neglected, the transmittance of the atmosphere is smaller than $1(\tau < 1)$, which leads to reduce the TB change $\Delta T = |T_{\text{target}} - T_{\text{sea}}|$ in (32). Thus, the atmosphere has a nonnegligible impact on the performance of the airborne PIMS for ship detection [35].

Numerical simulations are performed to the mentioned metallic ship in Section III-C, which is detected by an airborne PIMS with $\Delta\theta_X = \Delta\theta_Y = 0.5°$ and $\Delta T_B = 1\,K$. As shown in Fig. 5(a), the detection probability P_d with $P_f = 10^{-5}$ as a function of the detection distance L in the ideal case $\tau = 1$ is given in 10.7, 18.7, 23.8, 37.0, and 89.0 GHz. It is found that the detection distance is far in high frequency with the same detection probability. This results that the absolute difference

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

TB ($|T_{\text{target}}(\xi,\eta) - T_{\text{sea}}(\xi,\eta)|$) between the ship and the sea background is large in high frequency.

As shown in Fig. 5(b), the detection probability with $P_f = 10^{-5}$ as a function of the detection distance with 30-mm vapor is also given in 10.7, 18.7, 23.8, 37.0, and 89.0 GHz. It is found that there is a low detection probability in 23.8 and 89 GHz, in contrast to that in Fig. 5(a). It indicates that the vapor shows a high attenuation and has a large negative impact on the detection probability in high frequency. Meanwhile, there is a high attenuation in 23.8 GHz due to the fact that there is an atmospheric absorption band at 23.8 GHz [35].

As shown in Fig. 5(c), the detection probability with $P_f = 10^{-5}$ as a function of the detection distance with 0.3-mm water is given in 10.7, 18.7, 23.8, 37.0, and 89.0 GHz. It is found that the water shows a high attenuation in high frequency, and there is a low detection probability in 89 GHz, in contrast to that in Fig. 5(a).

As shown in Fig. 5(d) and (e), the detection probability with $P_f = 10^{-5}$ as a function of the detection distance with different vapor and water contents in 37 GHz is also given, respectively. The results show the higher the vapor and water contents, the lower the detection probability, as expected. Furthermore, the result indicates that the negative impact on the detection probability depends on the water and vapor contents. As a result, the bad weather with high vapor and water contents has a large negative impact on the detection performance of the airborne PIMS for ship detection, such as moderate rain and heavy rain.

B. Impact of the Sea Wind on PIMS

In the sea surface, sea wind can increase the sea surface roughness. Meanwhile, the sea surface is covered by foams in high wind speeds [36]. Both the sea surface roughness and the foams can increase the sea surface emissivity, which will lead to increase the sea surface TB. However, the TB is not almost changed by the wind for metallic ships. As a result, the wind will increase the TB absolute difference ($|T_{\text{target}}(\xi,\eta) - T_{\text{sea}}(\xi,\eta)|$) of the metallic ship and the sea in (29) that will further improve the detection performance of the airborne PIMS for metallic ship detection.

For the sea wind, numerical simulations are also performed to the mentioned metallic ship in Section III-C, which is also detected by the airborne PIMS with $\Delta\theta_X = \Delta\theta_Y = 0.5°$ and $\Delta T_B = 1\,K$. As shown in Fig. 5(f), the detection probability with $P_f = 10^{-5}$ as a function of the detection distance in 37 GHz is given in different wind speeds. It is found that the detection probability increases as the wind speed increases in the same detection distance. The results indicate that the sea wind can improve the detection performance of the PIMS for the metallic ship detection, as expected. As a result, the sea wind has a positive impact on the detection performance of the airborne PIMS for metallic ship detection.

V. Experiments, Results, and Discussion

To verify the practicability of ship detection by the airborne PIMS, a series of airborne experiments has been performed by an X-band PIMS, which is installed on an aircraft, as shown in Fig. 6 (left). The airborne experiments are performed in a sunny weather and a cloudy weather, respectively. The antenna array aperture of the X-band PIMS is synthesized from 19 antenna elements arranged in a Y-shaped configuration, as shown in Fig. 6 (middle). Each arm is six antenna elements, and an antenna element is in the hub. The −3-dB beamwidth of the antenna elements is about 65°. The main specifications of the X-band PIMS are given in Table I. The X-band PIMS is installed on the bottom of the aircraft, as shown in Fig. 6 (right). In addition, an optical camera is also installed on the bottom of the aircraft, which has the same field of views of the X-band PIMS and records the observation scenes in optical images for the data analysis.

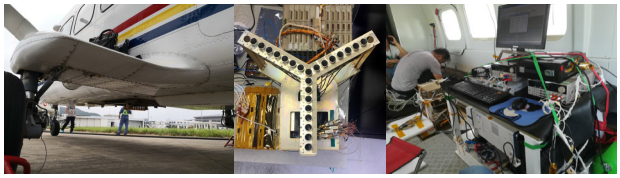


Fig. 6. (Left) Aircraft in the experiments. (Middle) X-band PIMS. (Right) X-band PIMS mounted on the aircraft.

TABLE I
Specifications of the X-Band PIMS

Specification	Parameter
Central Frequency	10.65GHz
Bandwidth	100MHz
Integration time	100ms/as needed
Antenna number	19
Adjacent spacing	0.95 λ

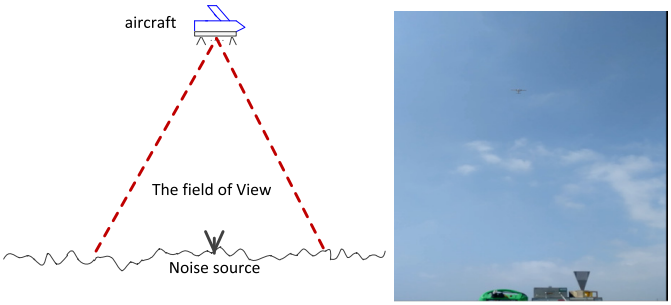


Fig. 7. (Left) Sketch of the external calibration method. (Right) Amplitude and phase errors calibrated by an external noise source located in the airline of the aircraft on the ground.

A. Data Processing Procedure

In the experiments, the external calibration method is performed, which only calibrate the amplitude and phase errors of the X-band PIMS [see Fig. 7 (left)]. As shown in Fig. 7 (right), an external noise source is located in the airline of the aircraft on the ground, which points toward the boresight of the X-band PIMS. In the experiments, the aircraft flies over the sea to randomly detect the ships by the X-band PIMS. For the X-band PIMS, the measured visibilities are calibrated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

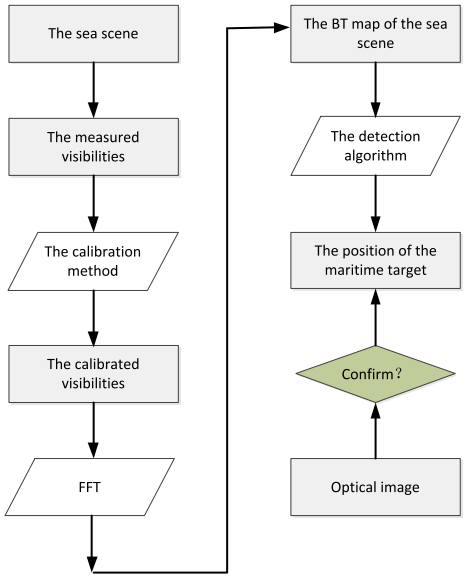


Fig. 8. Data processing procedure in the experiments.

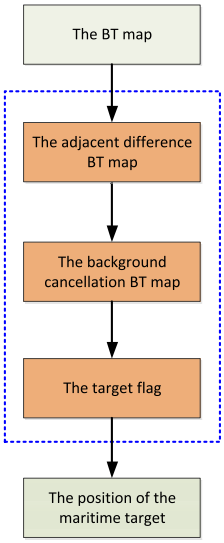


Fig. 9. Detection algorithm performed in the experiments.

by the external calibration method, which are recovered to the microwave TB maps using the fast Fourier transform (FFT) [37]. After that, a ship detection algorithm is performed to detect the ships. Finally, the ships detected by the ship detection algorithm are verified by the optical images. The processing procedure is given in Fig. 8. Due to the low angular resolution of the X-band PIMS of about 7.2° in the Blackman window, all TB maps of the scenes are retrieved by FFT with three-time zero-padded. In addition, the lines of the alias-free filed of views are not drawn in the TB maps.

In the experiments, the inputs of the ship detection algorithm are the original TB maps recovered by FFT with three-time zero-padded in the Blackman window, in which only the amplitude and phase errors are calibrated by the external calibration method. The ship detection algorithm used in the experiments is given in Fig. 9. The ship detection algorithm consists of three steps as follows.

Step 1 *(Adjacent Difference TB Map):* The adjacent difference TB maps are the difference of the adjacent TB maps. When a ship in the pixel (θ_0, φ_0) of the nth snapshot TB map or in the pixel (θ_1, φ_1) of the mth snapshot TB map, the microwave TB maps obtained by the airborne PIMS can be expressed as, respectively

$$\begin{aligned}\tilde{T}_B(\theta,\varphi,n) &= T_{\text{sea}}(\theta,\varphi,n) + (T_{\text{ship}}(\theta,\varphi,n) - T_{\text{sea}}(\theta,\varphi,n)) \\ &\quad \cdot\delta(\theta-\theta_0,\varphi-\varphi_0) + T_n(\theta,\varphi,n) \qquad (33)\\ \tilde{T}_B(\theta,\varphi,m) &= T_{\text{sea}}(\theta,\varphi,m) + (T_{\text{ship}}(\theta,\varphi,m) - T_{\text{sea}}(\theta,\varphi,m)) \\ &\quad \cdot\delta(\theta-\theta_1,\varphi-\varphi_1) + T_n(\theta,\varphi,m) \qquad (34)\end{aligned}$$

where n and m denote the serial number of the TB maps, $T_{\text{sea}}(\theta,\varphi,n)$ and $T_{\text{sea}}(\theta,\varphi,m)$ denote the sea surface TB maps, $T_{\text{ship}}(\theta,\varphi,n)$ and $T_{\text{ship}}(\theta,\varphi,m)$ denote the ship TB maps, and $T_n(\theta,\varphi,n)$ and $T_n(\theta,\varphi,m)$ denote the noise in the TB maps.

For the sea surface, the adjacent TB maps are considered to be almost the same

$$T_{\text{sea}}(\theta,\varphi,n) \approx T_{\text{sea}}(\theta,\varphi,m), \quad \text{when } |m-n| < L \qquad (35)$$

where L depends on the sea surface and the stability of the aircraft flight in the experiments.

Then, the adjacent TB difference map can be obtained by

$$\begin{aligned}&\Delta T(\theta,\varphi,n)\\ &= \tilde{T}_B(\theta,\varphi,n) - \tilde{T}_B(\theta,\varphi,m)\\ &= (T_{\text{ship}}(\theta,\varphi,n) - T_{\text{sea}}(\theta,\varphi,n))\\ &\quad \cdot\delta(\theta-\theta_0,\varphi-\varphi_0) - (T_{\text{ship}}(\theta,\varphi,m) - T_{\text{sea}}(\theta,\varphi,m))\\ &\quad \cdot\delta(\theta-\theta_1,\varphi-\varphi_1) + \sqrt{2}T_n(\theta,\varphi). \qquad (36)\end{aligned}$$

The TB flag is considered as

$$f(n) = \begin{cases} & \max\{\Delta T(\theta,\varphi,n)\} > N\cdot\sigma \\ 1, & \text{if and} \\ & \min\{\Delta T(\theta,\varphi,n)\} < -N\cdot\sigma \\ 0, & \text{else}\end{cases} \qquad (37)$$

in which "1" denotes that the ships exist in the snapshot TB map and "0" denotes that no ships exist in the snapshot TB map, N is a scaling threshold factor and a variable parameter, and σ is the standard deviation of the sea surface TB maps. When there are consecutive TB maps $(t_1,\cdots,t_l)$ flagged as "1," the ships are considered to exist in the TB maps.

Step 2 *(Background Cancellation TB Map):* In step 1, the TB maps flagged as "0" are considered to the sea surface TB maps

$$\tilde{T}_{\text{sea}}(\theta,\varphi,s) = T_{\text{sea}}(\theta,\varphi,s) + T_n(\theta,\varphi,s). \qquad (38)$$

The background cancellation TB maps are the difference between the TB maps of ships and the sea surface TB maps

$$\begin{aligned}\Delta T_s(\theta,\varphi,t) &= \tilde{T}_B(\theta,\varphi,t) - \tilde{T}_{\text{sea}}(\theta,\varphi,s_1)\\ &= (T_{\text{ship}}(\theta,\varphi,t) - T_{\text{sea}}(\theta,\varphi,s_1))\\ &\quad \cdot\delta(\theta-\theta_0,\varphi-\varphi_0) + \sqrt{2}T_n(\theta,\varphi) \qquad (39)\end{aligned}$$

in which $\tilde{T}_{\text{sea}}(\theta,\varphi,s_1)$ is the nearest adjacent TB map of the consecutive ship TB maps flagged as "1," which are flagged as "0" and considered as the sea background TB maps.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

Step 3 *(Target Flag Map):* As mentioned in Section II, the metallic ships are a low BT region in the PIMI and exhibit a "cold" TB characteristic in the PIMI. Based on the characteristic, the target flag maps are performed to flag the "low" TB regions in the background cancellation TB maps obtained by step 2, as "1," as follows:

$$\text{Flag}(\theta, \varphi, s) = \begin{cases} 1, & \Delta T_s(\theta, \varphi, s) \leq -M \cdot \sigma \\ 0, & \Delta T_s(\theta, \varphi, s) > -M \cdot \sigma \end{cases} \tag{40}$$

in which "1" denotes that there is a ship in the pixel and "0" denotes that there are no ships in the pixels, $s \in [t_1, \ldots, t_l]$ for one ship, and $M \cdot \sigma$ is a threshold, in which M is a scaling threshold factor and a variable parameter.

In the experiments, N and M are assigned to be 3. However, the parameters used in the ship detection algorithm can be adjusted as needed.

B. Experimental Results

In the experiments, the altitudes of the aircraft are 900, 1200, and 1800 m, respectively. In the three altitudes, the spatial resolutions of the X-band PIMS at boresight are about 110, 146, and 220 m, respectively.

One airborne experiment is performed in a sunny weather with the altitudes about 900 and 1200 m, in which the ships are detected many times by the airborne X-band PIMS using the ship detection algorithm. Four examples are shown in Fig. 10. As shown in Fig. 10(a), four original TB maps obtained by the airborne X-band PIMS are given, in which only the amplitude and phase errors are calibrated. It is found that the four original TB maps are not flat due to the fact the additive errors are not calibrated. In Fig. 10(a), a ship is found in each original TB map by step 1 in the ship detection algorithm. As shown in Fig. 10(b), the adjacent difference TB maps of the four examples with respect to Fig. 10(a) by step 1 in the ship detection algorithm are given with $n - m = 5$. As shown in Fig. 10(c), the background cancellation TB maps of the four examples with respect to Fig. 10(a) by step 2 in the ship detection algorithm are given, in which the opposite sea background TB maps are chosen as $\tilde{T}_{\text{sea}}(\theta, \varphi, s)$ in (39). It is found that there is a low TB region in each background cancellation TB map. As shown in Fig. 10(d), the target flag maps of the four examples with respect to Fig. 10(a) by step 3 in the ship detection algorithm are given. It is found that a ship is flagged in the target flag map. As shown in Fig. 10(e), the optical images of the four examples with respect to Fig. 10(a) are given to verify the flagged ships in the target flagged maps.

As shown in Fig. 10(c), there is a "low" TB region in each background cancellation TB map. As mentioned in Section II, the "low" TB region should be a metallic ship. Thus, the "low" TB region in the four background cancellation TB maps [see Fig. 10(c)] is flagged as "1" by step 3 in the ship detection algorithm, as shown in Fig. 10(d). As shown in Fig. 10(e), it is found, as expected, that there is indeed a metallic ship in each optical image, in which the location of the ship is the same as the "low" TB region in each background cancellation TB map [see Fig. 10(c)] and the pixels flagged as "1" in the target flag maps [see Fig. 10(e)].

It is found that the four examples in Fig. 10 are also flagged and detected in the consecutive TB maps by the ship detection algorithm. The consecutive background cancellation TB maps and the consecutive target flag maps of each example in Fig. 10 are also obtained. It is found that there is also a "low" TB region in the consecutive background cancellation TB maps of each example, which are flagged as "1" in the consecutive target flag maps. The "low" TB region in the consecutive background cancellation TB maps and the pixels flagged as "1" in the target flag maps are also demonstrated to be a ship by contrasting each background cancellation TB map and each target flag map to the corresponding optical image for each example. In addition, it is also found that the "low" TB region is always accompanied by a "high" TB region in Fig. 10(c), which is caused by ship wakes. The experimental results indicate that the ship detection algorithm can effectively detect the ships at seas and the ships can be accurately detected and consecutively tracked by the airborne X-band PIMS. It also means that the ships detected by the airborne PIMS are practicable in a sunny weather.

Another airborne experiment is also performed in a cloudy weather, in which the altitude of the aircraft is about 1800 m, as shown in Fig. 11 (left). Due to the cloud, the aircraft flies randomly over the sea. In this case, a metallic ship is also detected by the X-band PIMS. As shown in Fig. 11 (middle), two background cancellation TB maps about the ship are given, in which a similar "low" TB region and an accompanied "high" TB region also exist. The corresponding target flag maps about Fig. 11 (middle) are given in Fig. 11 (right). It is found that there is a target in Fig. 11 (right). Based on the conclusions drawn from Fig. 10, it should be a metallic ship. The detected ship in Fig. 11 is also analyzed and demonstrated by the consecutive background cancellation TB maps and the consecutive target flag maps. As a result, the results also exhibit that the metallic ships can be detected by the airborne X-band PIMS in a cloudy weather.

C. Results and Discussion

In the experiments, the integration time is 100 ms for ship detection. In this case, the sensitivity is about 1.5 K in the alias-free field of view of the airborne PIMS. The angular resolutions are about 7.2° in the x- and y-axes with the Blackman window. Thus, the detection factor Df_{system} of the X-band PIMS is estimated at about 42.2 $K^{-1}\text{rad}^{-2}$ by (11). In the experiments, the ships are painted metallic and exhibits about 0.99 emissivity in the X-band [19]. This leads to that the TB absolute difference between the painted metallic ships and the sea background is about 75 K at boresight estimated by the model [38]. The painted metallic ship in Fig. 10 is a tanker with about 230-m length and 32-m width, which are estimated by the optical images. According to (29), the detection probability P_d of the tanker at boresight as a function of the altitudes of the aircraft (the detection distance) with $P_f = 10^{-5}$ in ideal weather is simulated, which is shown as the black curve in Fig. 12. As shown in Fig. 12, it is found that the detection probability of the tanker is about 100% when

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

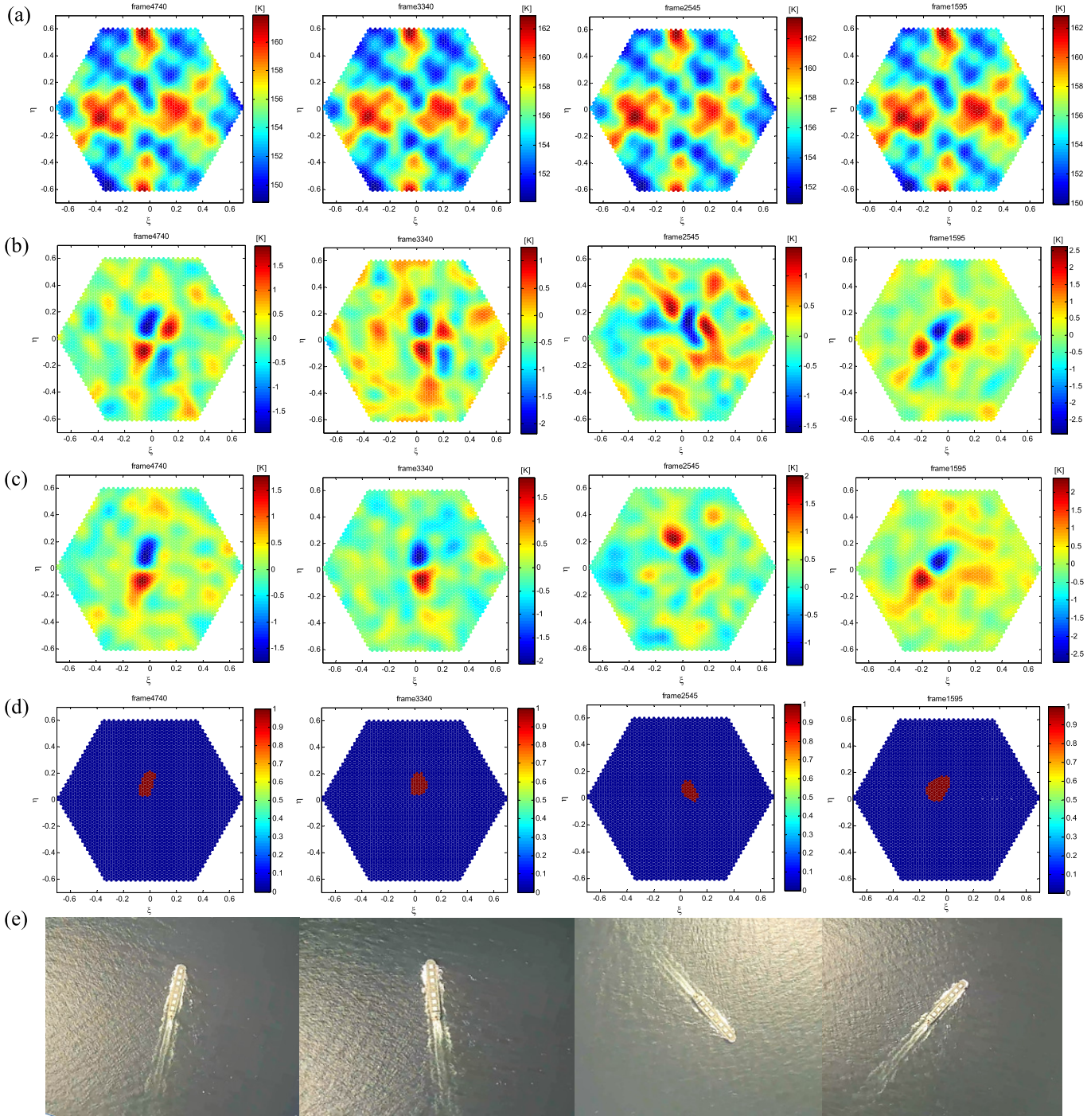


Fig. 10. Four examples detected by the X-band PIMS using the ship detection algorithm in the experiments. (a) Original TB maps of the four examples obtained by the airborne X-band PIMS. (b) Adjacent difference TB maps of the four examples by step 1 in the ship detection algorithm with $n - m = 5$. (c) Background cancellation TB maps of the four examples by step 2 in the ship detection algorithm. (d) Target flag maps of the four examples by step 3 in the ship detection algorithm. (e) Optical images of the four examples.

$L < 1900$ m, and the detection probability exceeds 90% when $L < 2250$ m.

In the experiments, the tanker, as shown in Fig. 13 (left), is detected by the X-band PIMS from 900 to 1200 m, respectively. In the two altitudes, the spatial resolution of the X-band PIMS is about 110 and 146 m, respectively. According to (6), the tanker in two altitudes is a line target with about two pixels in the PIMI. The weather is sunny, and the attenuation of the atmosphere between the aircraft and the tanker can be neglected. According to (6) and (12), the target detection factor Df_{target} for the tanker is 3.07 and 4.11 $K^{-1}\text{rad}^{-2}$, respectively. According to (29), the tanker can be detected with $P_d = 100\%$ by the X-band PIMS with $P_f = 10^{-5}$ from 900 to 1200 m, as the black circles in Fig. 12. In the experiments, the speed of the aircraft is about 60 m/s and the integration time is 100 ms. As a result, the tanker in the alias-free field

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

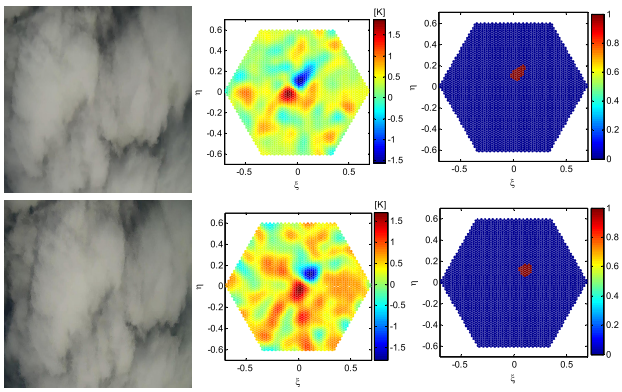


Fig. 11. (Left) Optical images in a cloudy weather. (Middle) Background cancellation TB maps. (Right) Corresponding target flag maps in the experiments.

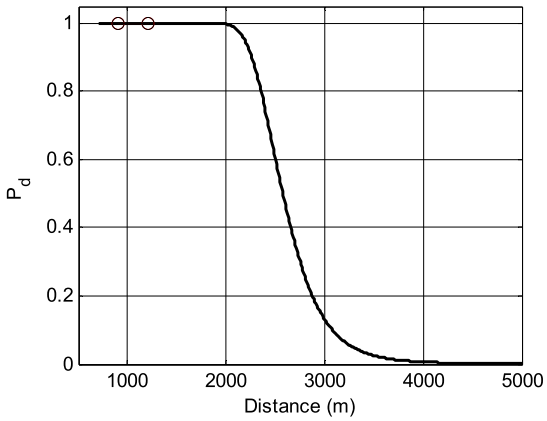


Fig. 12. Detection probability of the tanker at boresight as a function of the detection distance of the X-band PIMS when $P_f = 10^{-5}$.



Fig. 13. Optical image of (Left) tanker and (Right) container ship detected by the airborne X-band PIMS in the experiments.

of views of the X-band PIMS is about 80 snapshots from 900 m and about 107 snapshots from 1200 m, respectively. The ship detection probability in the experiments can also be defined as the ratio of the ship detected snapshots, in which the ships are detected by the airborne X-band PIMS, to the total snapshots in the alias-free filed of views. The statistics of the detection probability of the tanker by the airborne X-band PIMS are given in Table II. It is found that the detection probability of the tanker about 100% well agrees with the theoretical detection probability, as the black curve shown in Fig. 12. A container ship, as shown in Fig. 13 (right), is also detected by the X-band PIMS from 900 m. The statistics of the detection probability of the container ship are also given in Table II. The detection probability of the container ship in

TABLE II

STATISTICS OF THE SHIPS DETECTED BY THE AIRBORNE X-BAND PIMS IN THE EXPERIMENTS

Target	Altitude (m)	Time	p_D
Tanker	900	14:22:49	100%
Tanker	900	14:25:33	100%
Tanker	900	14:27:25	100%
Tanker	900	14:29:09	100%
Tanker	900	14:31:54	100%
Tanker	900	14:34:34	100%
Tanker	900	14:37:16	100%
Tanker	900	14:39:39	100%
Container ship	900	14:42:55	100%
Container ship	900	14:44:11	100%
Tanker	900	14:47:52	100%
Tanker	1200	14:53:40	100%

the experiments also well agrees with the theoretical detection probability.

It must be pointed out that the metallic ships detected by the X-band PIMS in the experiments can also be detected by an active microwave sensor, such as SARs. However, the ships with small radar signatures can also be detected by the airborne PIMS due to the high BT characteristic against the sea background in the PIMI, such as rigid inflatable boats, wooden boats, rubber boats, and fiberglass boats [19], [28]. In addition, the ship wake is a significant signature for the moving ships and submerged crafts with a high BT characteristic in the PIMIs, which can be considered as an important characteristic for ship detection and submerged craft detection by the airborne PIMS.

VI. CONCLUSION

Ship detection is important for a wide range of applications. In this article, an airborne PIMS is proposed as a powerful complementary tool for ship detection with a key advantage under low visibility conditions. The fundamental of ship detection by the airborne PIMS is introduced in terms of the contrast in the passive microwave images. The detection factor is proposed to quantitatively assess the performance of the airborne PIMS for ship detection, which is defined as the reciprocal of the product of the sensitivity and the angular resolution of the airborne PIMS. The larger the detection factor, the better the detection performance of the airborne PIMS. On the other hand, the detection probability based on the CFAR is also addressed, which depends on the FAR, the sensitivity and angular resolutions of the airborne PIMS, the microwave radiation cross section of the ships, and the TB absolute difference between the ships and the sea background. The feasibility of ship detection by the airborne PIMS is also demonstrated by numerical simulations. The impacts of the atmosphere and the sea wind on the detection performance of the airborne PIMS for ship detection are also addressed in terms of theoretical analyses and numerical simulations. The results show that bad weather with high vapor and water contents degrades the detection performance of the airborne

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

PIMS for ship detection, such as moderate rain and heavy rain. However, the results also show that the sea wind has a positive impact on the detection performance of the airborne PIMS for metallic ship detection.

To verify the practicability of ship detection by the airborne PIMSs, a series of airborne experiments has been performed by an X-band PIMS in a sunny weather and a cloudy weather. The data processing procedure and the ship detection algorithm are introduced. The experimental results demonstrate the practicability of ship detection by the airborne PIMS in a sunny weather and a cloudy weather. The experimental results also indicate that the ship detection algorithm can effectively detect and track the ships by the X-band PIMS. In addition, it is also found that the "low" TB region caused by the ships in the PIMI is always accompanied by the "high" TB region, which is caused by the ship wakes. The detection probability of the ships in the experiments is also analyzed and it well agrees with the theoretical detection probability, as expected. In addition, the ships with small radar signatures can also be detected by the airborne PIMS, such as rigid inflatable boats, wooden boats, rubber boats, and fiberglass boats. As a conclusion, the airborne PIMS can be considered as a powerful complementary tool for ship detection.

As mentioned in [19] and [28], the airborne PIMS has a variety of applications with a key advantage under low visibility conditions. In addition, as mentioned in [19] and [28], the ship wakes also show a significant signature in the PIMI from space, and the signature of the ship wake will be considered as a significant signature for ship detection in further works.

ACKNOWLEDGMENT

The authors would like to thank the anonymous reviewers for their valuable comments and suggestions, which help to improve the quality of this article.

REFERENCES

[1] Z. Liu, H. Wang, H. Weng, and L. Yang, "Ship rotated bounding box space for ship extraction from high-resolution optical satellite images with complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 8, pp. 1074–1078, Aug. 2016.

[2] F. Yang, Q. Xu, and B. Li, "Ship detection from optical satellite images based on saliency segmentation and structure-LBP feature," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 602–606, May 2017.

[3] S. Brusch, S. Lehner, T. Fritz, M. Soccorsi, A. Soloviev, and B. van Schie, "Ship surveillance with TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 1092–1103, Mar. 2011.

[4] Y. Wang and H. Liu, "A hierarchical ship detection scheme for high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 4173–4184, Oct. 2012.

[5] H. J. Roarty, E. R. Lemus, E. Handel, S. M. Glenn, D. E. Barrick, and J. Isaacson, "Performance evaluation of SeaSonde high-frequency radar for vessel detection," *Marine Tech. Soc. J.*, vol. 45, no. 3, pp. 14–24, May 2011.

[6] S. Grosdidier, A. Baussard, and A. Khenchaf, "HFSW radar model: Simulation and measurement," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 9, pp. 3539–3549, Sep. 2010.

[7] Y. Wang, X. Mao, J. Zhang, and Y. Ji, "Detection of vessel targets in sea clutter using *in situ* sea state measurements with HFSWR," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 302–306, Feb. 2018.

[8] G. K. Høye, T. Eriksen, B. J. Meland, and B. T. Narheim, "Space-based AIS for global maritime traffic monitoring," *Acta Astronautica*, vol. 62, nos. 2–3, pp. 240–245, Feb. 2008.

[9] M. D. Graziano, M. D'Errico, and E. Razzano, "Constellation analysis of an integrated AIS/remote sensing spaceborne system for ship detection," *Adv. Space Res.*, vol. 50, no. 3, pp. 351–362, Aug. 2012.

[10] C. Zhu, H. Zhou, R. Wang, and J. Guo, "A novel hierarchical method of ship detection from spaceborne optical image based on shape and texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 9, pp. 3446–3456, Sep. 2010.

[11] J.-W. Lu, Y.-J. He, H.-Y. Li, and F.-L. Lu, "Detecting small target of ship at sea by infrared image," in *Proc. IEEE Int. Conf. Automat. Sci. Eng. CASE*, Shanghai, China, Oct. 2006, pp. 165–169.

[12] H. Wang, Z. Zou, Z. Shi, and B. Li, "Detecting ship targets in spaceborne infrared image based on modeling radiation anomalies," *Infr. Phys. Technol.*, vol. 85, pp. 141–146, Sep. 2017.

[13] Y. Ji, J. Zhang, J. Meng, and Y. Wang, "Point association analysis of vessel target detection with SAR, HFSWR and AIS," *Acta Oceanologica Sinica*, vol. 33, no. 9, pp. 73–81, Sep. 2014.

[14] S. Park, C. J. Cho, B. Ku, S. Lee, and H. Ko, "Compact HF surface wave radar data generating simulator for ship detection and tracking," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 969–973, Jun. 2017.

[15] S. Maresca, P. Braca, J. Horstmann, and R. Grasso, "Maritime surveillance using multiple high-frequency surface-wave radars," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 5056–5071, Aug. 2014.

[16] F. Mazzarella, M. Vespe, and C. Santamaria, "SAR ship detection and self-reporting data fusion based on traffic knowledge," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 8, pp. 1685–1689, Aug. 2015.

[17] J. Tang, C. Deng, G. B. Huang, and B. Zhao, "Compressed-domain ship detection on spaceborne optical image using deep neural network and extreme learning machine," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1174–1185, Mar. 2015.

[18] R. Appleby, D. G. Gleed, R. N. Anderton, and A. H. Lettington, "High-performance passive millimeter-wave imaging," *Opt. Eng.*, vol. 32, no. 6, pp. 1370–1373, May 1993.

[19] L. Yujiri, M. Shoucri, and P. Moffa, "Passive millimeter-wave imaging," *IEEE Microw. Mag.*, vol. 4, no. 3, pp. 39–50, Sep. 2003.

[20] Y. Cui, J. Yang, Y. Yamaguchi, G. Singh, S.-E. Park, and H. Kobayashi, "On semiparametric clutter estimation for ship detection in synthetic aperture radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 3170–3180, May 2013.

[21] G. Yang, B. Li, S. Ji, F. Gao, and Q. Xu, "Ship detection from optical satellite images based on sea surface analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 3, pp. 641–645, Mar. 2014.

[22] A. R. Harvey and R. Appleby, "Passive mm-wave imaging from UAVs using aperture synthesis," *Aeronaut. J.*, vol. 107, no. 1068, pp. 87–98, Feb. 2003.

[23] Y. A. Pirogov, "Passive millimeter-wave imaging," *Radiophys. Quantum Electron.*, vol. 46, nos. 8–9, pp. 594–603, Aug./Sep. 2003.

[24] H.-L. Lu *et al.*, "Research on the passive detection technology using space-borne synthesis aperture microwave radiometers for the sea surface target," *J. Infr. Millim. Waves*, vol. 38, no. 5, pp. 674–682, Oct. 2019.

[25] C. S. Ruf, C. T. Swift, A. B. Tanner, and D. M. Le Vine, "Interferometric synthetic aperture microwave radiometry for the remote sensing of the Earth," *IEEE Trans. Geosci. Remote Sens.*, vol. GRS-26, no. 5, pp. 597–611, Sep. 1988.

[26] M. Peichl, H. Suss, and S. Dill, "High-resolution passive millimeter-wave imaging technologies for reconnaissance and surveillance," *Proc. SPIE*, vol. 5077, pp. 77–86, Aug. 2003.

[27] L. Hailiang, Q. Li, L. Yan, Y. Li, and H. Li, "Low-level radio-frequency interference detection algorithm based on European centre for medium-range weather forecasting for soil moisture and ocean salinity," *J. Appl. Remote Sens.*, vol. 9, no. 1, 2015, Art. no. 095996.

[28] N. A. Salmon, "Outdoor passive millimeter-wave imaging: Phenomenology and scene simulation," *IEEE Trans. Antennas Propag.*, vol. 66, no. 2, pp. 897–908, Feb. 2018.

[29] F. T. Ulaby, R. K. Moore, and A. K. Fung, *Microwave Remote Sensing: Active and Passive: Microwave Remote Sensing Fundamentals and Radiometry*, vol. 1. Reading, MA, USA: Addison-Wesley, 1981.

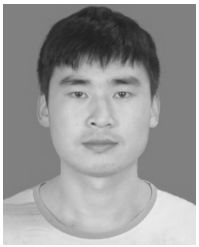
[30] M. Martín-Neira *et al.*, "SMOS instrument performance and calibration after six years in orbit," *Remote Sens. Environ.*, vol. 180, pp. 19–39, Jul. 2016.

[31] D. Zhu, F. Hu, F. He, L. Wu, J. Li, and C. H. Jang, "A novel optimal design of antenna array in aperture synthesis radiometers," *IEEE Antennas Wireless Propag. Lett.*, vol. 14, pp. 1227–1230, 2015.

[32] D. Zhu, F. Hu, L. Lang, P. Tang, X. Peng, and F. He, "Radiometric sensitivity and angular resolution optimization of thinned linear arrays in microwave interferometric radiometry," *IEEE Trans. Antennas Propag.*, vol. 67, no. 1, pp. 568–573, Jan. 2019.

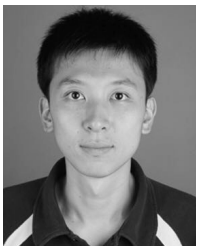
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.

[33] D. Zhu, F. Hu, X. Peng, and H. Lu, "On baseline weighting for radiometric sensitivity analysis in microwave interferometric radiometry," *IEEE Antennas Wireless Propag. Lett.*, vol. 18, no. 4, pp. 791–795, Apr. 2019.
[34] A. Camps, I. Corbella, J. Bará, and F. Torres, "Radiometric sensitivity computation in aperture synthesis interferometric radiometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 2, pp. 680–685, Mar. 1998.
[35] F. J. Wentz and T. Meissner, "Atmospheric absorption model for dry air and water vapor at microwave frequencies below 100 GHz derived from spaceborne radiometer observations," *Radio Sci.*, vol. 51, no. 5, pp. 381–391, May 2016.
[36] T. Meissner and F. J. Wentz, "The emissivity of the ocean surface between 6 and 90 GHz Over a large range of wind speeds and Earth incidence angles," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 8, pp. 3004–3026, Aug. 2012.
[37] E. Anterrieu and A. Khazaal, "Brightness temperature map reconstruction from dual-polarimetric visibilities in synthetic aperture imaging radiometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 3, pp. 606–612, Mar. 2008.
[38] L. Klein and C. T. Swift, "An improved model for the dielectric constant of sea water at microwave frequencies," *IEEE J. Ocean. Eng.*, vol. OE-2, no. 1, pp. 104–111, Jan. 1977.



Hailiang Lu received the B.S. degree in electrical engineering from Three Gorges University, Yichang, China, in 2009, the M.S. degree in electromagnetic field and microwave technology from the Guilin University of Electronic Technology, Guilin, China, in 2012, and the Ph.D. degree in electromagnetic field and microwave technology from the Huazhong University of Science and Technology, Wuhan, China, in 2016.

He is currently a Senior Engineer with the China Academy of Space Technology (Xi'an), Xi'an, China. His research interests include the passive microwave remote sensing, the passive microwave detection, and the radio-frequency interference detection.



Yinan Li received the B.S. and M.S. degrees in electrical engineering from Xidian University, Xi'an, China, in 2007 and 2010, respectively.

He is currently a Senior Engineer with the China Academy of Space Technology (Xi'an), Xi'an. His research interests include the passive microwave remote sensing and the microwave radiometer design.



Hao Li received the B.S. degree in electrical engineering from Xidian University, Xi'an, China, in 2003, and the M.S. degree from the China Academy of Space Technology (Xi'an), Xi'an, China, in 2006.

He is currently a Professor with the China Academy of Space Technology (Xi'an). His research interests include the passive microwave remote sensing and the microwave radiometer design.



Rongchuan Lv received the B.S. and M.S. degrees in electrical engineering from Xidian University, Xi'an, China, in 2004 and 2007, respectively.

She is currently a Professor with the China Academy of Space Technology (Xi'an), Xi'an. Her research interests include the passive microwave remote sensing and the microwave radiometer design.



Liang Lang received the B.S. degree in electrical engineering from South Central University for Nationalities, Wuhan, China, in 1997, and the M.S. and Ph.D. degrees in electromagnetic field and microwave technology from the Huazhong University of Science and Technology, Wuhan, in 1999 and 2009, respectively.

He is currently an Associate Professor with the School of Electronic Information and Communications, Huazhong University of Science and Technology. The focus of his research was passive microwave remote sensing, electromagnetic simulations, and related research.



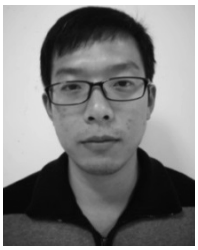
Qingxia Li received the B.S., M.S., and Ph.D. degrees from the Department of Electronics and Information, Huazhong University of Science and Technology, Wuhan, China, in 1987, 1990, and 1999, respectively.

He is currently a Professor with the School of Electronic Information and Communications (Science and Technology on Multi-spectral Information Processing Laboratory), Huazhong University of Science and Technology. His research interests include microwave remote sensing, antenna array, signal processing, and wireless communication.



Guangnan Song received the B.S. and M.S. degrees in electrical engineering from the Nanjing University of Science and Technology, Nanjing, China, in 2010 and 2013, respectively.

He is currently a Senior Engineer with the China Academy of Space Technology (Xi'an), Xi'an, China. His research interests include the passive microwave remote sensing and the microwave radiometer design.



Pengfei Li received the B.S. degree in electrical engineering from the Harbin Institute of Technology (HIT), Weihai, China, in 2012, and the M.S. degree in electrical engineering from the China Academy of Space Technology (Xi'an), Xi'an, China, in 2015.

He is currently an Engineer with the China Academy of Space Technology (Xi'an). His research interests include microwave remote sensing, radiometer system design, and system calibration.



Ke Wang received the B.S. degree in applied physics from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1995.

He is currently a Senior Engineer with the China Academy of Space Technology (Xi'an), Xi'an, China. His research interests include microwave remote sensing, radar system design, and radar signal processing.

Liang Xue, photograph and biography not available at the time of publication.



Dong Zhu (M'18) received the B.S. degree in communication engineering and the Ph.D. degree in electronic science and technology from the Huazhong University of Science and Technology, Wuhan, China, in 2013 and 2018, respectively.

He is currently a Post-Doctoral Fellow with the Department of Electronic Engineering, Tsinghua University, Beijing, China. His research interests include antenna arrays, array signal processing, microwave remote sensing, and numerical optimization techniques for electromagnetic problems.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:59:16 UTC from IEEE Xplore. Restrictions apply.